

SmartLogistics — Intelligent Supply Chain & Delivery Orchestration Platform [Part - B]

TransFleet is building **SmartLogistics**, an intelligent large-scale logistics platform designed to support modern e-commerce businesses, retail chains, warehouses, third-party logistics providers, and enterprise supply chain operations.

The company is experiencing rapid growth in shipment volume, warehouse operations, courier partners, and delivery requests, which has exposed several limitations in their current systems:

- Shipment creation and dispatch workflows are slow and partially manual, delaying order fulfillment.
- Inventory data across warehouses and fulfillment centers is inconsistent.
- Courier assignment and route planning are inefficient during peak delivery windows.
- Shipment tracking events are delayed, duplicated, or lost between systems.
- High operational traffic results in delays when processing large dispatch batches and status updates.
- Logistics interactions generate valuable operational data, but it is not being leveraged for optimization or decision-making.

To address these challenges, TransFleet is commissioning a new backend for SmartLogistics with the following business goals.

TransFleet now wants to evolve SmartLogistics into an **intelligent logistics platform powered by GenAI**.

Description / Problem Statement

This phase focuses on solving operational inefficiencies and decision-making challenges using GenAI.

Current issues include:

- Courier assignment is rule-based and not adaptive
- Delays are detected late with limited proactive action
- Route optimization requires manual intervention
- Operational teams lack quick access to insights
- Logistics data is underused for forecasting and recommendations

This phase introduces:

- AI logistics assistant
- Intelligent route and dispatch suggestions
- Delay prediction support
- Natural language operational insights
- Streaming AI responses for operations teams

Business Goals & Vision

Intelligent Logistics Operations

Operations users should be able to ask:

- Which shipments are likely to be delayed today?
- Why is warehouse throughput low this week?
- Suggest best courier for urgent deliveries
- Show failed deliveries by region

The platform should provide contextual answers.

Operational Productivity Enhancement

Teams should be able to generate:

- Daily dispatch summaries
- Delay reports
- Courier performance summaries
- Warehouse bottleneck insights
- Operational recommendations

Smooth Real-Time Interactions

Long-running AI responses should support streaming or asynchronous delivery and must not block core logistics traffic.

Core Functional Requirements

Content / Data Preparation Workflow

When shipment and operational data is updated:

- Relevant data should be chunked
- Embeddings should be generated
- Searchable vector records should be stored
- Data should become available for semantic retrieval

Intelligent Logistics Assistant

A. Contextual Operations Q&A

Users ask logistics-related questions.

The system should:

- Retrieve relevant shipment / warehouse / courier data
- Use contextual information
- Generate meaningful responses
- Handle vague or incomplete operational queries

B. Optimization & Recommendation Support

Generate:

- Courier assignment suggestions
- Delay mitigation recommendations
- Warehouse load balancing suggestions
- Dispatch prioritization insights

Analytics Metrics

- AI Assistant Usage
- Questions Asked / Answered
- Delay Prediction Accuracy
- Optimization Recommendation Usage
- Average AI Response Time
- Operational Productivity Impact

Observability

Ability to diagnose failures in:

- AI assistant interactions
- Retrieval pipeline
- Streaming responses
- External LLM provider calls

Expected Outcomes

- Intelligent logistics assistant
- Better operational decision-making
- Context-aware shipment insights
- Automated reporting workflows
- Scalable AI processing layer
- PRD Requirement
 - AI use-cases
 - Functional / non-functional requirements
 - Delivery milestones
 - Feature traceability

Tech Stack

- LangGraph / LangChain
- LLM Provider (OpenAI / Groq / Anthropic)
- Vector DB